

Experiment 7: Random Sampling and Sampling Distribution.

Aim:

To study and random sampling and understand the sampling distribution of sample means using python.

Description:

A population is generated and random samples of different sizes are selected .The sample means are calculated and plotted to observe how the sampling distribution changes with sample size.

Code:

```
import numpy as np

import matplotlib.pyplot as plt

population_mean = 75

population_std = 12

population_size = 40000

population = np.random.normal(population_mean, population_std, population_size)

sample_sizes = [15, 40, 80]

num_samples = 300

sample_means = {}

for size in sample_sizes:

    sample_means[size] = []

    for _ in range(num_samples):

        sample = np.random.choice(population, size=size, replace=False)

        sample_means[size].append(np.mean(sample))

plt.figure(figsize=(12, 8))

for i, size in enumerate(sample_sizes):

    plt.subplot(len(sample_sizes), 1, i + 1)

    plt.hist(sample_means[size], bins=25, alpha=0.7,

            label=f'Sample Size {size}')

    plt.axvline(np.mean(population), color='red',
```

```

linestyle='dashed', linewidth=1.5,

label='Population Mean')

plt.title(f'Sampling Distribution (Sample Size {size})')

plt.xlabel('Sample Mean')

plt.ylabel('Frequency')

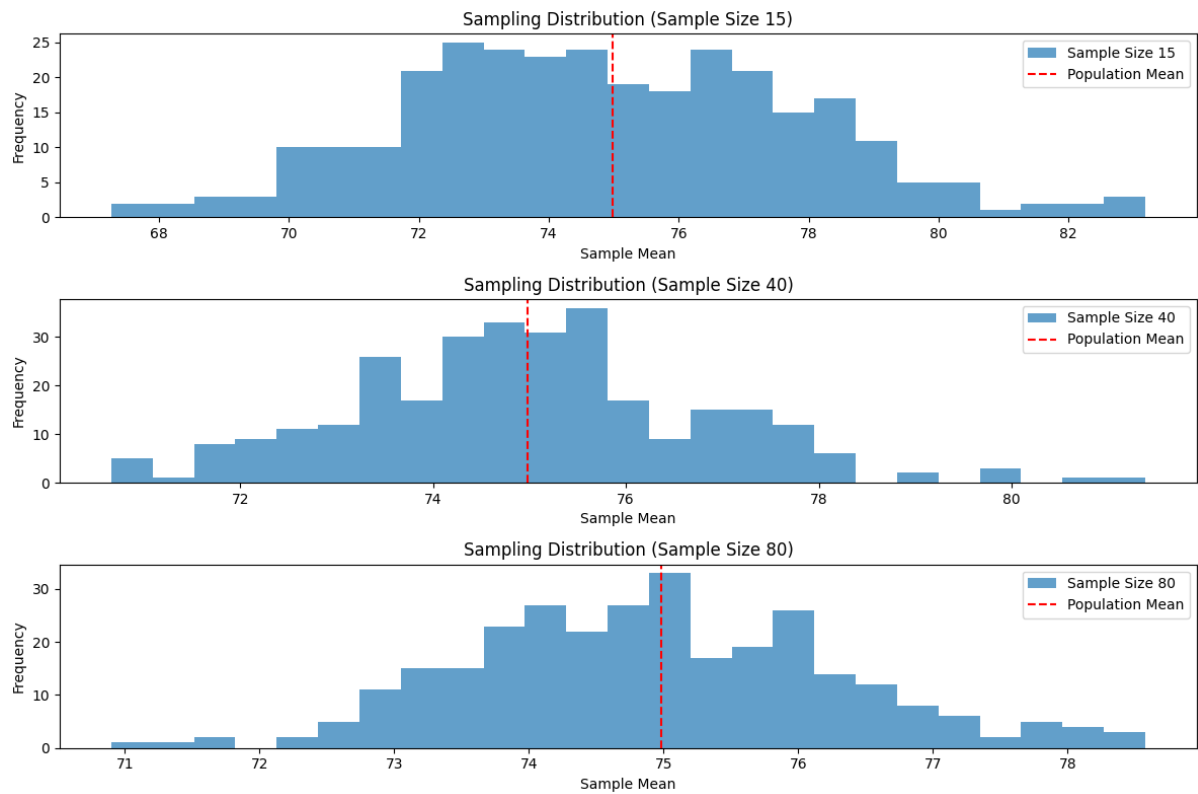
plt.legend()

plt.tight_layout()

plt.show()

```

Output:



Result:

Random sampling and sampling distributions were successfully obtained and analysed.